SWS3009

SWS3009A
Deep Learning

SWS3009B
Robotics

LECTURE 1
COMBINED BRIEFING

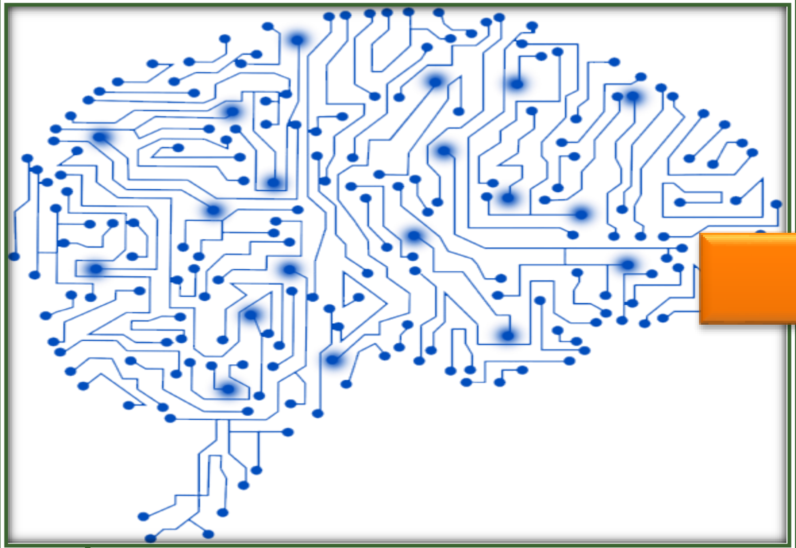
Outline

- **Preliminary**
  - Course Overview
  - Schedule Overview

- **Team Forming** (20 minutes)

- **Team Ice Breaking + Brainstorming** (20 minutes)

Course Overview



Data Intelligence

2-3 Students

Mobility Platform

1-2 Students

Course Objective (Brief)

Engineer an intelligent mobile platform

Knowledge	**Skill**

Theory	**Hands-on**	**Problem Solving**	**Teamwork**

Session Overview

Combined Briefing
9th July 9:30 am

SWS3009A Session I
9th July 10:30 am to 5 pm

SWS3009A Session II
10th July 9:00 am to 5 pm

SWS3009A Session III
13th July 9:30 am to 5 pm

SWS3009A Session IV
14th July 9:30 am to 5 pm

SWS3009B Session I
9th July 10:30 am to 5 pm

SWS3009B Session II
10th July 9:30 am to 5 pm

SWS3009B Session III
13th July 9:30 am to 5 pm

SWS3009B Session IV
14th July 9:30 am to 5 pm

Project Schedule Overview

Baseline Model
15th – 16th July

↓

Treasure Hunt
17th July

↓

Advanced Model
18th July to 26th July

↓

Final Demonstration
27th July

↓

Showcase
29th July

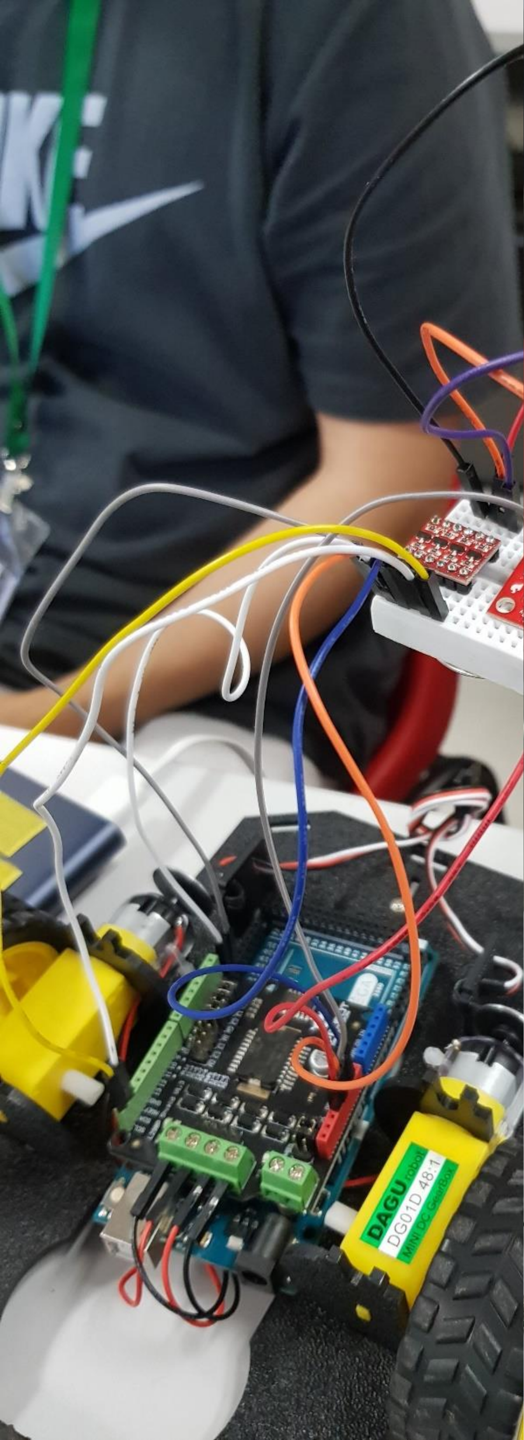
A STORY OF TWO COURSES

1 + 1 > 2 ?

- **Baseline Model** (17^{th} July):
  - Remote controlled vehicle with camera
  - Ability to classify images
  - → **"Treasure Hunt" Robot!**

- **Advanced Model** (27^{th} July):
  - **Your imagination is the limit**

  - We will provide components, consultation, suggestions and evaluation to **push you to the limit!**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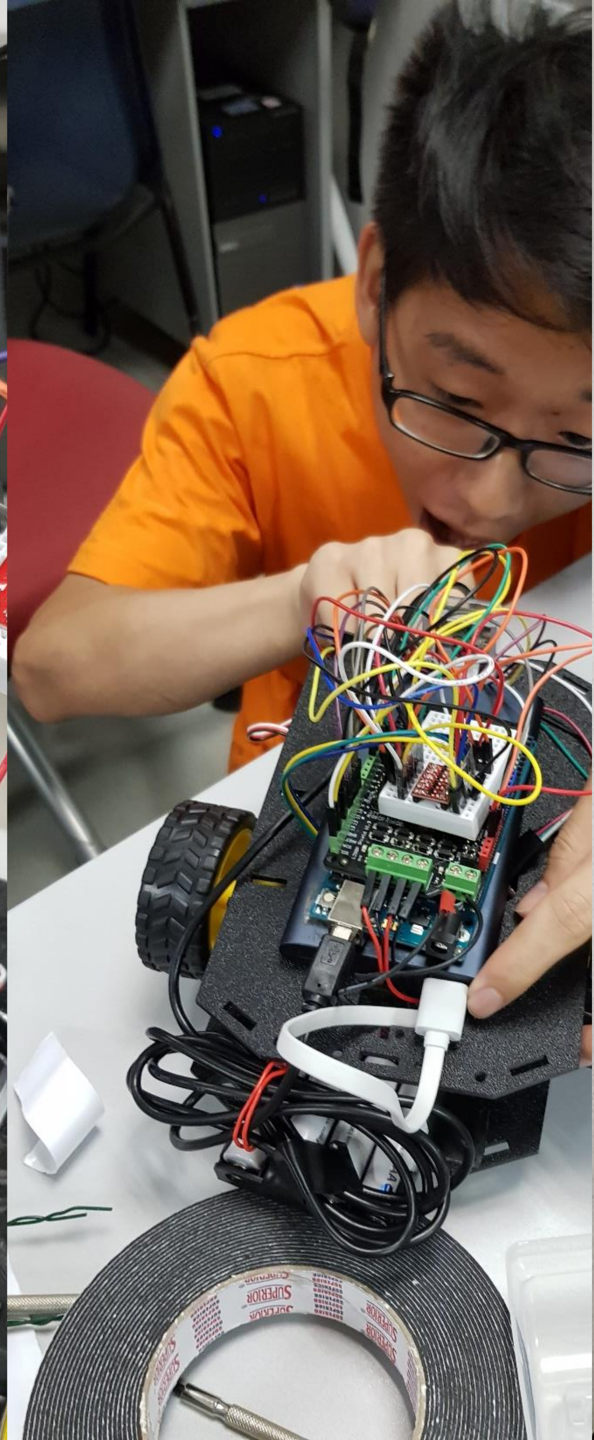
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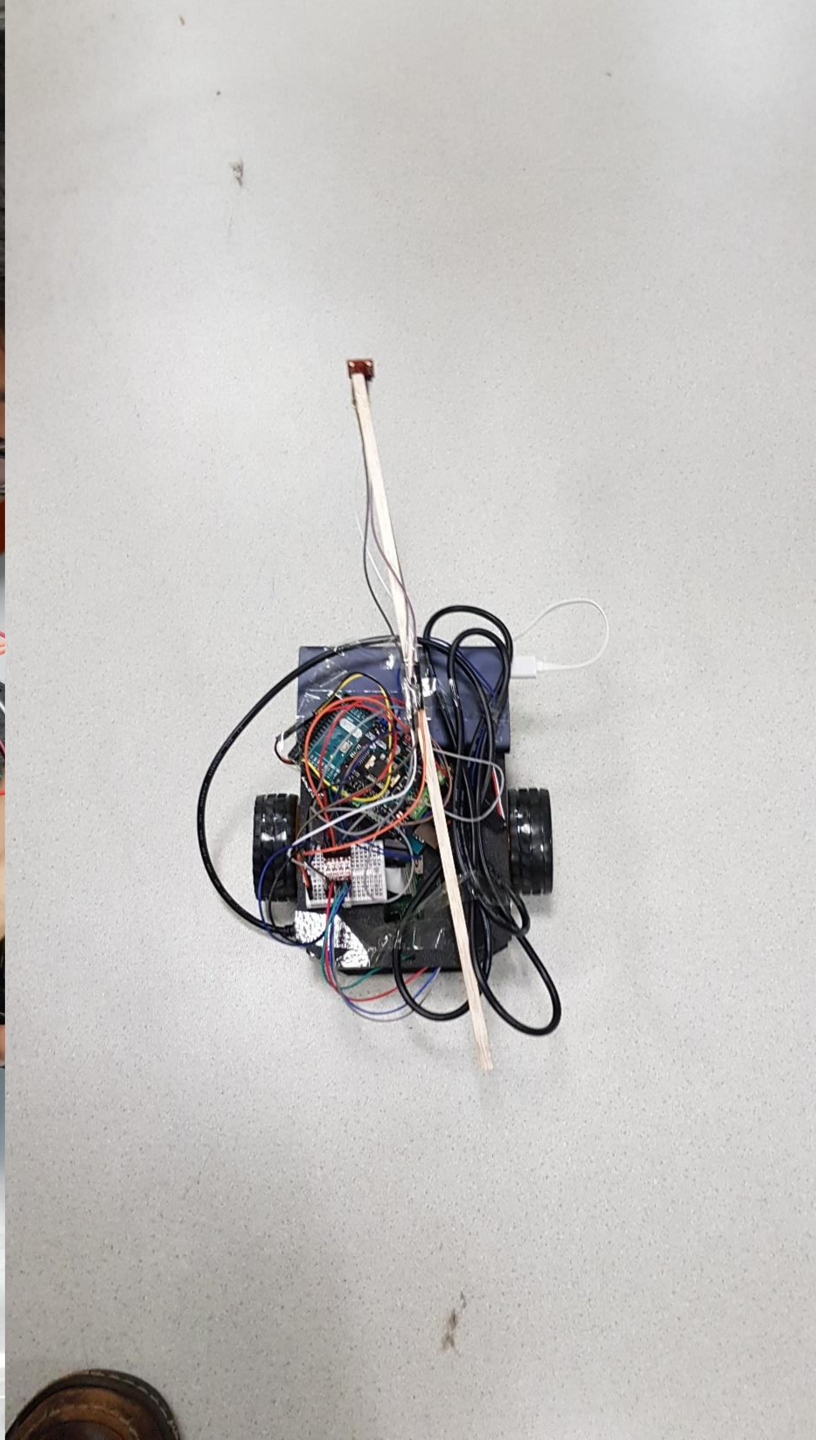
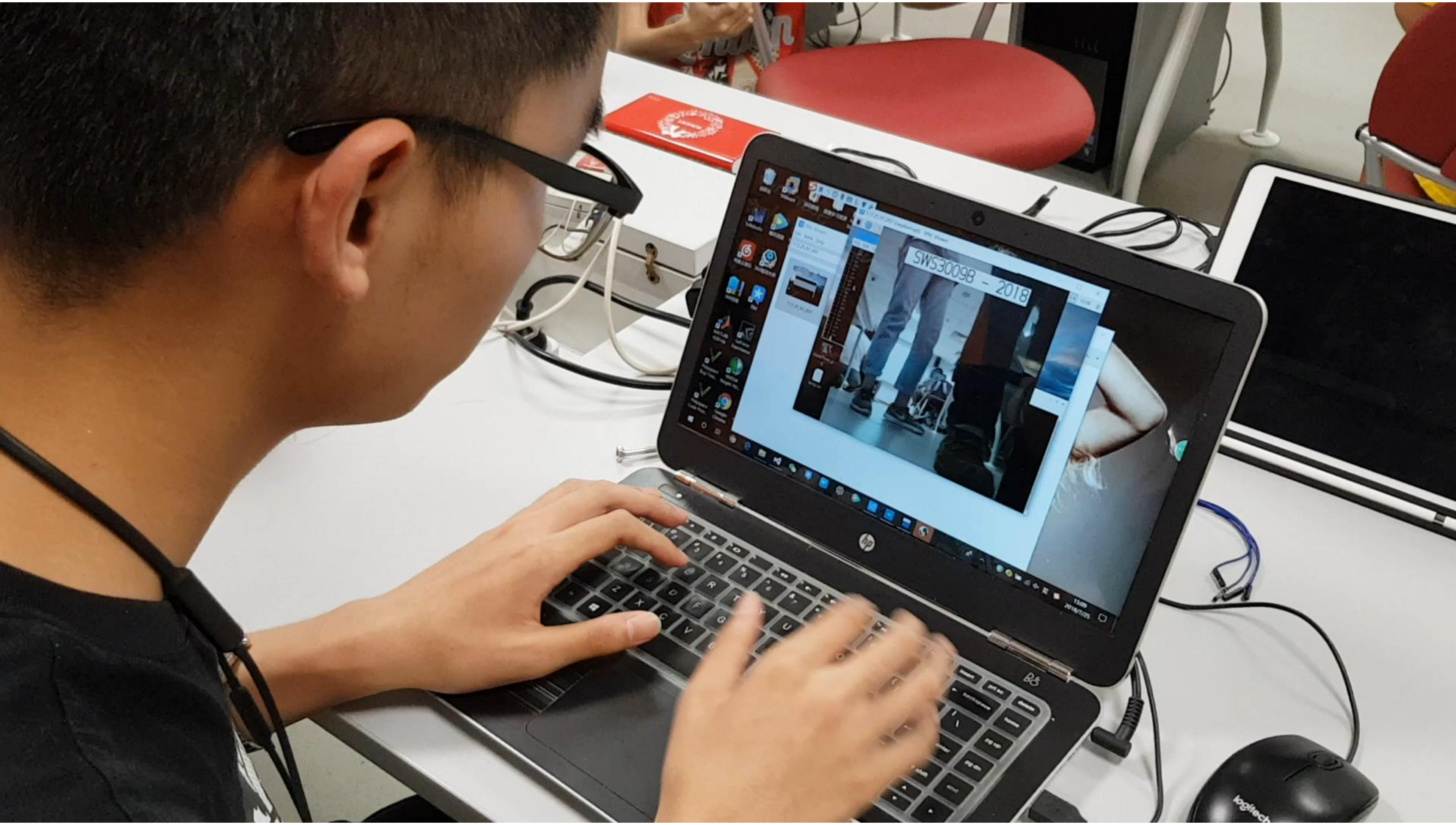
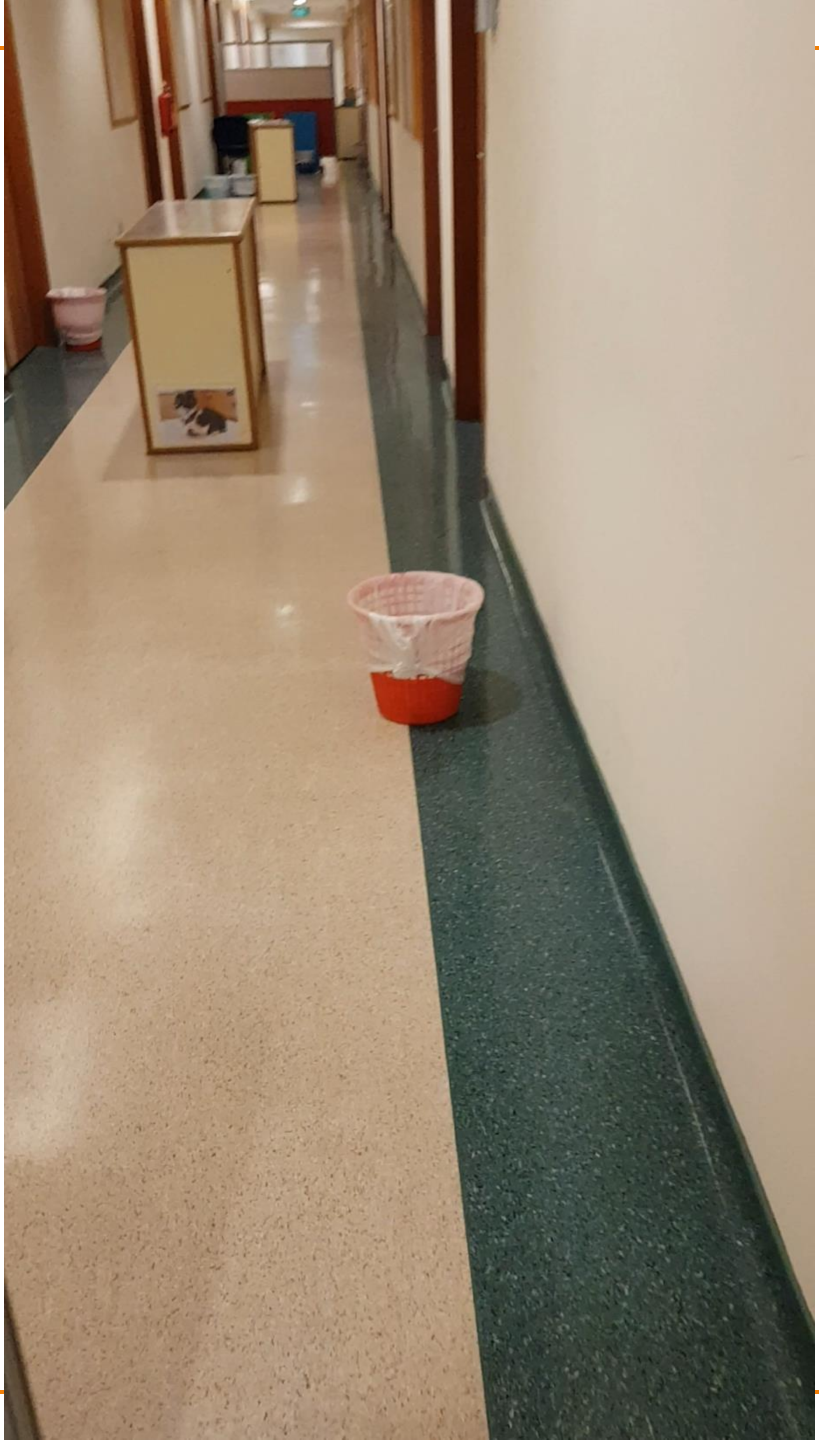
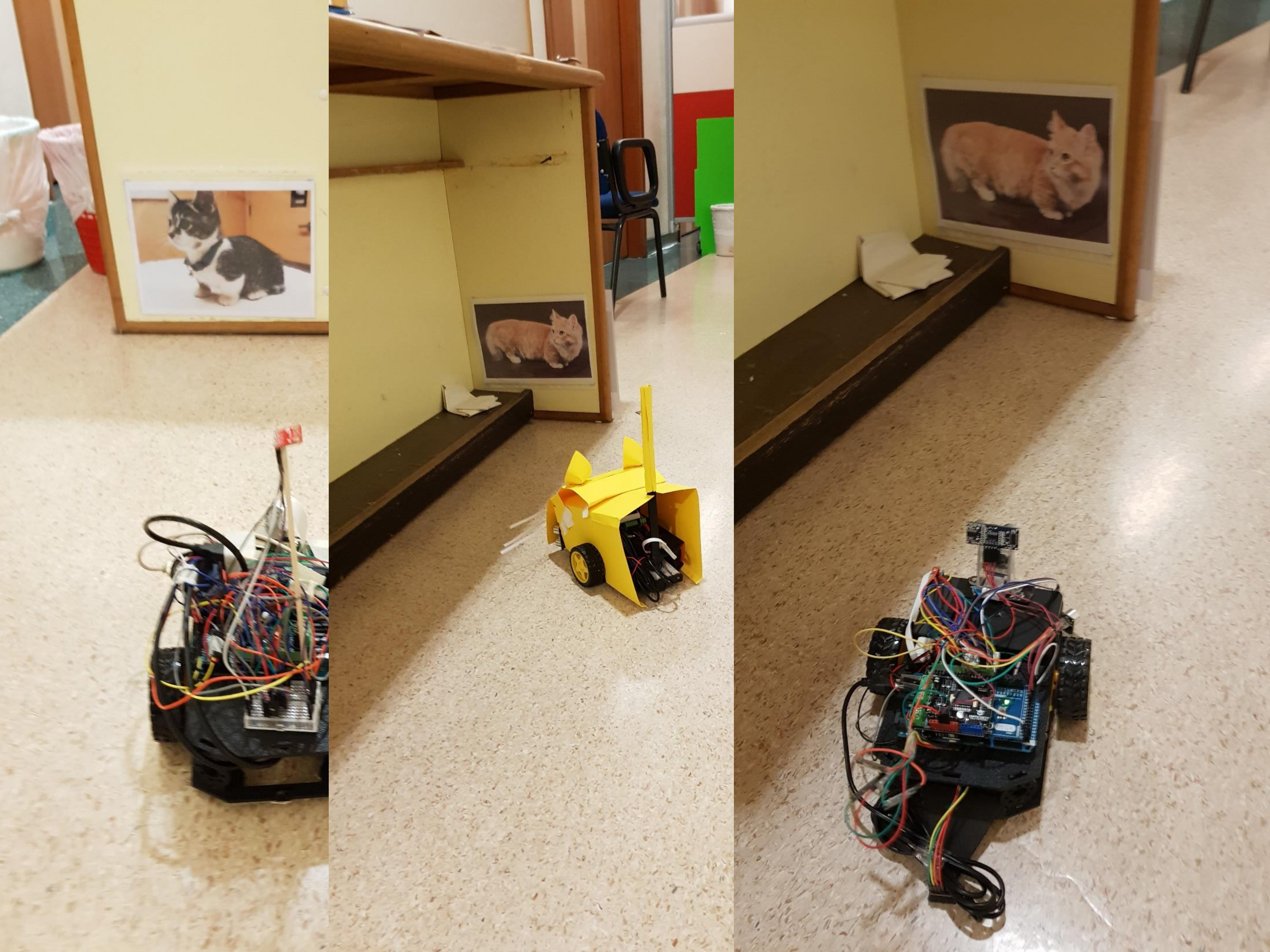


Remote Driving – Stealth Car! ☺







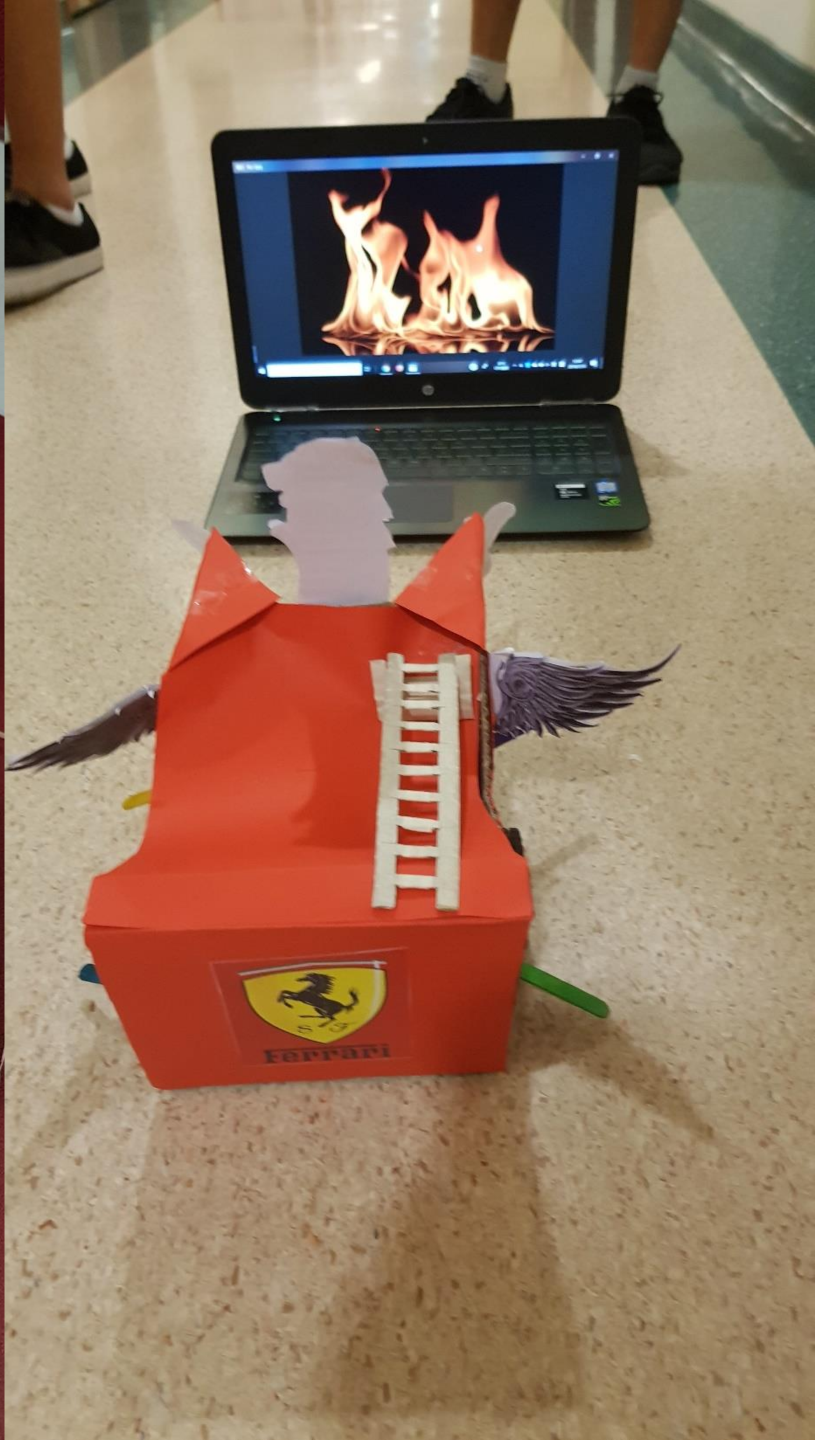
Behind the Scenes – Control Room

Advanced Model Examples

- **Mobility** + **Intelligence** = ???

- Some possibilities:
  - **Autonomous / Remote controlled explorer**
    - Map out area
    - Take photo of locations
    - Identify / flag the locations (e.g. office occupier)
  - **Search and rescue bot**
    - Explore area to look for survivor (humanoid or otherwise)
  - **Semi-autonomous explorer**
    - Navigate via "signpost" along the path etc.


Team 8


Ferrari

Be creative!

Crazy difficult and fun!

Build real things!

- Highlight of SWS3009:
  - You will build a **real platform** interacting with **real world**
  - Pick up practical know-how (tensor flow, electronics etc)
  - You can decide how crazy / fun the final deliverable

LET'S START WORKING!

Team Forming (3-4 Persons)

- A **balanced** team:
  - 2-3 Deep Learning Experts
  - 1-2 Robotic Experts
  - **Different Universities**

TEAM FORMATION

- We will now call you team-by-team. When you hear your name being called, stand up.
- Locate your other team mates, and move to a part of the LT where you can discuss.
- Please take note and memorize your team number (e.g. SWS3009-03)

SWS3009-1

XU TIANRUN
HAO ZIQUAN
PAN KUO
XU ZIQI
WU JUNCHAO

SWS3009-2

CHEN ZIYAN
YU TAO
ZHANG YUANYE
WEI JIACHENG
PENG SICHENG

SWS3009-3

ZHANG ZIYU
NIU CHUYUAN
ZHENG QIAOWEN
CHEN XUANHAO
HUA YARUO

SWS3009-4

WU YONGHONG
XU CHENGYI
YU YIHANG
ZHANG RONGCHAO
WU SHIHAO

SWS3009-5

GLADYSHEV ILIAN
SUN MINGCHEN
CHEN HAN
ZHAO YI
WANG ZHUANGXU

SWS3009-6

WANG ZHIYAO
SUN MINGLIANG
TAO ZUI
ZHANG CHENKAI
BIN JIARUI

SWS3009-7

ZHOU ZIMO
WU LINYANG
HUANG ZHUOYAN
HE QIUSHI
ZHANG SIKAI

SWS3009-8

ZHANG ZHIWEN
ZHOU WENYUE
TONG YI
XU ZIMO
YE ZILE

SWS3009-9

WU QINGYA
FAN TONGYU
DU MINGMIN
WU YI
WU FEIYAO

SWS3009-10

YU ZHIPING
ZHANG RUNZE
CAI CHENYI
LUO TIANRUI

SWS3009-11

CHEN ZENGCHUN
GU SONGTAO
WANG TIANQI
LIN ZIQI

SWS3009-12

ZHAO BOXIONG
WANG LECHEN
ZHANG XUHONG
XIN YIXUAN

SWS3009-13

JIN DAJUN
JI TUO
CURRY JOHN

SWS3009-14

WANG ZIYU
WU GAOYU
WAN JIAWEN

SWS3009-15

GVOZDENOVIC DUNJA
YANG SICHANG
LI YIQI

Discussion – 10 Minutes

- Meet your group mates?
- Where are you all from?
- Come up with a cool team name!

Baseline Model

- Remote Controlled Robotic Vehicle

- Sensors to provide environment information:
  - Ultrasonic, Compass, etc

- Camera to:
  - View the surrounding (for driving)
  - Capture image for classification

- Deep learning models to classify images

Discussion – 20 Minutes

- What's your **advanced model**?

- Improvements on both fronts:
  - **Deep Learning**: Classification of different type of information? Better classification?

  - **Robotic**: Improved control? Autonomous? Build map (with images)?

  - **Combined**: Deep learning driven autonomous vehicle?

Discussions

- Identify **KEY problems** to solve
- Suggest 1-2 solutions for each problem
  - High level description only

Let's GO!

- SWS3009A – Deep Learning:
  - Stay Here
- SWS3009B – Robotics
  - Please go to Makers @ SoC
- We will start at 10:XX (to be confirmed)

END